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Metro Rail Scheduler - Demo Guide

Ye guide aapko project ko run karne aur demo dikhane mein help karegi. Ismein step-by-step instructions hain taaki aapka presentation smooth rahe.

1. Project Ko Run Kaise Karein (Setup)

Sabse pehle project ko apne local machine par chalane ke liye ye steps follow karein:

Prerequisites

- **Node.js** (v18 ya usse upar) install hona chahiye.
- **Terminal** (PowerShell ya CMD) open karein.

Installation Steps

1. Root Dependencies Install Karein:

```
npm install
```

2. Backend Dependencies Install Karein:

```
cd backend  
npm install
```

Agar `.env` file nahi hai, to `.env.example` ko copy karke `.env` banayein.

3. Wapas Root Par Jayein:

```
cd ..
```

Running the App

Ab backend aur frontend dono ko ek saath start karne ke liye ye command run karein:

```
npm run dev:all
```

- **Frontend:** `http://localhost:5173` par chalega.
- **Backend:** `http://localhost:4000` par chalega.



2. Demo Flow (Kya Dikhana Hai?)

Demo ke waqt ye main features dikhayein:

Step 1: Login & Authentication

- **Staff Login:** Admin portal access karne ke liye Staff login dikhayein.
- **Passenger Signup:** Naye users kaise register karte hain, vo dikhayein.
- *Tip: Mock data already included hai, to direct login bhi kar sakte hain.*

Step 2: Main Dashboard (Staff View)

- **Fleet Health:** Dikhao ki kitni trains active hain aur kitni maintenance mein hain.
- **Station Analytics:** Footfall aur crowd management ke graphs (Recharts use kiye gaye hain).
- **Recent Alerts:** Koi bhi system alerts ya delays.

Step 3: Admin Portal - Schedule Management

- **View Schedules:** Purani schedules ki list dikhayein.
- **Add/Edit Schedule:** Ek naya train schedule create karke dikhayein (CRUD functionality).
- **Delete Schedule:** Kis tarah se purani schedules remove ki jati hain.

Step 4: Live Tracking & Maps (Leaflet)

- **Interactive Map:** Mumbai Metro ke routes aur live train positions dikhayein.
- **Station Info:** Map par station markers par click karke details dikhayein.

Step 5: Passenger Information System (PIS)

- **Timetables:** Passengers ko train timings kaise dikhti hain.
- **Route Finder:** Ek station se dusre station ka rasta kaise search hota hai.

3. Troubleshooting (Agar Kuch Na Chale)

- **Port Busy Error:** Agar port already use mein hai, to `npm run dev:fresh` use karein. Ye purane processes ko kill karke naya server start karega.
 - **Backend Connection:** Check karein ki backend terminal mein koi error to nahi hai. Database connection ya env files check karein.
 - **Node Version:** Hamesha latest LTS version use karein.
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Pro Tips for Demo

- **Responsive Design:** Browser window ko chhota karke dikhayein ki app mobile par bhi achhi lagti hai.
 - **Dark Mode:** Agar toggle hai, to dark mode switch karke premium feel dikhayein.
 - **Realistic Data:** Presentation ke waqt mention karein ki data real-time scenario ke hisaab se realistic rakha gaya hai.
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All the best for your demo! 🚀